

VPA IB DP CS — Year 1 Semester 1 Assessment Pack

Year 1 Semester 1 Assessment Pack 评估包 Pínggū Bāo

Topics: 1 (System Fundamentals) + 2 (Computer Organization) | Level: SL + HL

1. Assessment Philosophy

All assessments are aligned to the IB DP Computer Science Guide (First Assessment 2024) and use IB command terms, markbands, and question styles. Formative assessments build toward summative mastery. EAL students receive bilingual keyword lists and extended time (if approved by IB coordinator).

2. Semester 1 Assessment Calendar

Week	Assessment	Type	Duration	Marks	IB Alignment
1	Diagnostic Test	Baseline	40 min	30	IGCSE recap + IB readiness
5	Topic 1 Progress Check	Formative	40 min	25	Paper 1 short-answer
9	Topic 1 End-of-Topic Test	Summative	1 hour	50	Paper 1 style (4, 8, 12-mark Qs)
14	Topic 2 Progress Check	Formative	40 min	30	Paper 1 calculations + diagrams
16	IA Client Proposal	IA Milestone	Homework	Pass/Fail	Criteria A: Planning
18	Semester Examination	Summative	SL 1h30 / HL 2h10	SL 70 / HL 100	Paper 1 Mock

3. Paper 1 Question Types and Mark Schemes

Question Type A: Define / State (1 – 2 marks)

Example: Define the term "volatile memory." [1]

Markscheme: Memory that loses its contents when power is removed. Accept: RAM is volatile. Do not accept: "temporary memory" without qualification.

Question Type B: Describe (3 – 5 marks)

Example: Describe the role of the control unit (CU) in the CPU. [3]

Markscheme:

Award [1] for stating it controls/fetches/decodes/executes instructions

Award [1] for describing it sends control signals to ALU/registers/bus

Award [1] for explaining it coordinates timing via clock signals

Question Type C: Explain (5 – 8 marks)

Example: Explain why cache memory improves CPU performance. [6]

Markscheme (IB-style markbands):

Marks	Descriptor
1 – 2	Basic identification that cache is faster than RAM. Limited explanation.
3 – 4	Clear explanation of temporal/spatial locality. Some reference to speed hierarchy.
5 – 6	Comprehensive explanation including locality principles, speed comparison (cache vs RAM vs disk), reduced bus traffic, and impact on FDE cycle timing.

Question Type D: Evaluate / Discuss (8 – 12 marks)

Example: Evaluate the use of parallel changeover versus direct changeover when implementing a new school information system. [10]

Markscheme (IB markbands for essays):

Marks	Descriptor
1 – 3	Limited analysis. One-sided or descriptive only. Few relevant points.
4 – 6	Some analysis with both sides presented. Examples given but not fully developed. Basic conclusion.
7 – 8	Good analysis with balanced argument. Relevant examples from context. Clear conclusion with justified judgement.
9 – 10	Excellent critical analysis. Sophisticated understanding of context. Well-chosen examples. Convincing, nuanced conclusion.

4. Topic 1 Sample Assessment Questions

Q1 (a) — Define [1]

Define the term stakeholder. (利益相关者 lìyì xiāngguān zhě)

Award [1] for: A person or group with an interest in or affected by a system.

Q1 (b) — Describe [3]

Describe three components of an information system.

Award [1] each for any three of: hardware, software, data, procedures, people. [1] each for valid description.

Q2 — Explain [6]

Explain the purpose of a feasibility study before system development begins.

Markband:

[1–2]: Identifies one type of feasibility (technical/economic/operational)

[3–4]: Explains two types with relevant examples; shows why each matters

[5–6]: Comprehensive coverage of all three types with clear links to project risk reduction and informed decision-making

Q3 — Construct [4]

A school library has the following entities: BOOK, MEMBER, LOAN. A member can borrow many books. A book can be borrowed by many members (over time). Each loan records one member and one book.

Construct an ER diagram showing these entities, relationships, and cardinality.

Award [1] for three entities; [1] for correct relationships; [1] for M:N between MEMBER and BOOK; [1] for LOAN as associative entity (or alternative valid representation).

Q4 — Evaluate [10]

A hospital is replacing its 20-year-old patient record system. Evaluate the use of phased changeover for this implementation.

5. Topic 2 Sample Assessment Questions

Q5 — State / Define [2]

- (a) State the purpose of the program counter (PC). [1]
- (b) State the purpose of the memory address register (MAR). [1]

Q6 — Describe [4]

Describe the fetch stage of the fetch-decode-execute cycle.

Award [1] for: PC contents copied to MAR; [1] for: Address placed on address bus; [1] for: Memory returns instruction to MDR; [1] for: PC incremented.

Q7 — Explain [6]

Explain how virtual memory allows a computer to run programs that require more RAM than is physically installed.

Q8 — Calculate / Show [5]

- (a) Convert the denary number 75 to 8-bit binary. [2]
- (b) Represent $\square 75$ using 8-bit two's complement. Show your working. [3]

Q9 — Construct [4]

A security system sounds an alarm ($A = 1$) when either motion is detected ($M = 1$) AND the system is armed ($S = 1$), OR when a manual panic button is pressed ($P = 1$) regardless of whether the system is armed.

- (a) Construct a truth table for this system. [2]
- (b) Draw a logic circuit for this system. [2]

Q10 — Discuss [12]

Discuss the impact of increasing the word length of a CPU on computer performance and system design.

6. Bilingual Glossary for Assessments

All students receive a bilingual keyword list before each assessment. EAL students may use an approved glossary in examinations (subject to IB regulations).

7. IA Client Proposal — Marking Criteria

Criterion	Requirement	Pass Criteria
A.1 Client Identification	Real, specific, contactable client	Name, role, organisation, contact method provided
A.2 Problem Definition	Clear, genuine problem statement	Problem described in client's own words; evidence of consultation
A.3 Rationale	Why this problem matters	At least two reasons why solution is needed
A.4 Success Criteria	Measurable, testable criteria	3-5 criteria, each with measurement method

8. Assessment for Learning (AfL) Strategies

Peer marking: Students mark sample responses using IB markschemes before attempting their own

DIRT time: Dedicated lesson time after each assessment for reflection and improvement

Bronze/Silver/Gold targets: Students set differentiated improvement goals

Exit tickets: 3-minute written reflection at end of every lesson

Retrieval practice: Weekly low-stakes quizzes on prior content

Assessment Pack Checklist:

- ☐ Diagnostic test prepared and photocopied
- ☐ All formative assessments have markschemes
- ☐ Summative assessments mirror IB Paper 1 timing and structure
- ☐ Bilingual glossaries prepared for EAL students
- ☐ IA proposal criteria shared with students
- ☐ Past paper questions compiled by topic
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Assessment Pack